

Fullstack Developer — Prescreening Questions

Part A — Candidate Profile

P1 — Work style & independence

Describe how you typically work when given a task with minimal direction. Do you prefer to figure things out independently, or do you seek regular check-ins and guidance? Give a specific example from a past role.

P2 — Startup environment fit

Have you worked in a startup or small team before? If yes, describe what that looked like — team size, your responsibilities, and how the pace/culture differed from larger organizations. If no, what makes you interested in joining one now?

P3 — Aspirations

Where do you see yourself in 2–3 years? What kind of work, responsibilities, or impact are you working toward?

P4 — Motivation for a new role

What is prompting you to look for a new opportunity right now? What are you specifically looking for in your next position that you are not getting in your current or most recent role?

P5 — Expected salary

What is your expected gross monthly salary?

Part B — Technical Questions

Q1 — AI-Powered WhatsApp Order Processing (No Token Costs)

Topics: AI/ML · Gemma / Ollama · Python · Intent extraction

Question

You are given a dataset of WhatsApp chat messages used by customers to purchase goods (e.g. “I’d like 3 bags of cement and 2 tins of paint please”). The business wants an AI-powered auto-reply system that extracts order intent and responds naturally — but cannot use token-billed APIs (OpenAI, Gemini API, etc.).

- a)** Describe your end-to-end architecture for running a model like Gemma 3 (or similar open-weight LLM) locally or on a self-hosted server to handle these messages.
 - b)** How would you structure the prompt / context so the model reliably extracts: item name, quantity, unit, and customer intent (order / inquiry / complaint)?
 - c)** How do you test and measure the accuracy of the extraction? Describe a concrete evaluation methodology including metrics you would track.
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Q2 — SLA Analytics Dashboard for Multi-Step Approval Workflows

Topics: Next.js · PostgreSQL · Recharts / Tremor · Data modeling

Question

You have an approval workflow system. Each workflow has multiple steps; each step records `assigned_at` and `completed_at` timestamps. Example: “Manager A — assigned 14:05, approved 17:06 → 181 min elapsed.”

- a)** Design the PostgreSQL schema to store workflows, steps, assignees, and SLA durations in a way that supports flexible querying (by department, step type, date range, user).
 - b)** What Next.js architecture choices (App Router, React Server Components, API routes, a UI component library like Tremor or shadcn) would give you high configurability with minimal custom code for the dashboard?
 - c)** List the specific charts and tables you would include in the dashboard and explain what each one reveals to a business analyst reviewing process bottlenecks.
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Q3 — Testing & Monitoring Google Cloud Tasks Scheduled Workflows

Topics: Google Cloud Tasks · Cloud Scheduler · Cloud Logging · Python

Question

You are implementing scheduled background workflows using Google Cloud Scheduler → Cloud Tasks → a Python HTTP worker. Schedules may trigger daily or on-demand.

- a) How do you test Cloud Tasks locally during development — without deploying to GCP every time?
 - b) Once deployed, how do you monitor that tasks were triggered, executed, and completed successfully? What GCP services or third-party tools do you use?
 - c) How do you handle and alert on task failures, retries, and dead-letter scenarios?
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Q4 — PostgreSQL Query Performance & Schema Optimization

Topics: PostgreSQL · Indexing · Query optimization

Question

You have a transactions table with 10 million rows. A query that filters by `user_id`, `status`, and a date range on `created_at` is running slow (4+ seconds).

- a) Walk through how you would diagnose this using EXPLAIN (ANALYZE, BUFFERS) — what would you look for?
 - b) What index strategy would you apply, and why? Consider composite vs partial indexes.
 - c) Are there any schema or query rewrite changes that could help beyond indexing?
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Q5 — Next.js API Design: Auth, Rate Limiting & Error Handling

Topics: Next.js · App Router · Middleware · Security

Question

You are building a Next.js 14 App Router application that exposes REST API routes consumed by both the internal frontend and external webhook callers.

- a) How do you implement JWT-based authentication in App Router API routes? Where does verification happen — middleware vs route handler?

- b)** How would you implement per-user rate limiting without a dedicated Redis instance (or with one if preferred)?
 - c)** Describe your pattern for consistent, typed error responses across all API routes.
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Q6 — Python Async Workers & Background Job Patterns

Topics: Python · FastAPI · asyncio · Background jobs

Question

Your Python FastAPI service needs to handle long-running tasks (e.g. generating a PDF report, calling a slow third-party API) triggered by an HTTP request, without blocking the response.

- a)** What patterns do you use? Compare: BackgroundTasks in FastAPI, Celery, ARQ (async Redis queue), and Cloud Tasks HTTP workers. When would you choose each?
 - b)** How do you report task progress back to the frontend (e.g. “report 60% complete”)?
 - c)** How do you ensure a task is not run twice if the HTTP request is retried?
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Q7 — Fullstack Integration: End-to-End Data Flow & Deployment

Topics: System design · Docker · CI/CD · Cloud Run

Question

Sketch the architecture for a system where: (1) a Next.js frontend allows users to submit orders, (2) a Python FastAPI backend processes them and stores results in PostgreSQL, (3) a nightly Cloud Tasks job generates a summary report sent via email.

- a)** Draw or describe the component diagram — what talks to what, over which protocol?
 - b)** How do you manage secrets and environment config across Next.js (Vercel) and Python (Cloud Run)?
 - c)** Describe a minimal CI/CD pipeline (GitHub Actions) that lints, tests, and deploys both services on merge to main.
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